

Repair Request - ER Diagram

Logical Model + Performance Boundary Review

Document Control	Value
Document ID	RR-ERD-001
Version	1.2 - Performance Review Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Supersedes	RR-ERD-001 v1.1
Revision Scope	Review high-growth relationships and correct notification timestamp boundary without adding entities or changing cardinality.
Approval	Business Owner / Project Manager + Architecture/QA review - pending

Revision rule / กติกาการแก้ไข: เอกสารฉบับนี้เป็นการปิด Audit ด้าน Performance/Technical Design เท่านั้น ไม่เปลี่ยน State, Role, Guard, SLA, Permission, Notification Recipient หรือ Business Semantics ที่ล็อกไว้เดิม หากคำตัดสินด้าน Performance ใดไม่มีอยู่ใน Business Baseline จะระบุเป็น **Gap / Proposed Technical Decision / Pending Approval** และไม่ถือเป็น Requirement เดิมโดยอัตโนมัติ.

Document control note: หน้านี้เป็น Document Control ที่มีผลสำหรับ revision นี้และ supersede ค่า Version/Status บนหน้าปกของ source baseline ที่ถูกรักษาไว้ในหน้าถัดไปเพื่อ traceability.

Repair Request - ER Diagram

Logical ER Model + Physical SQL Server Mapping / Baseline Revision

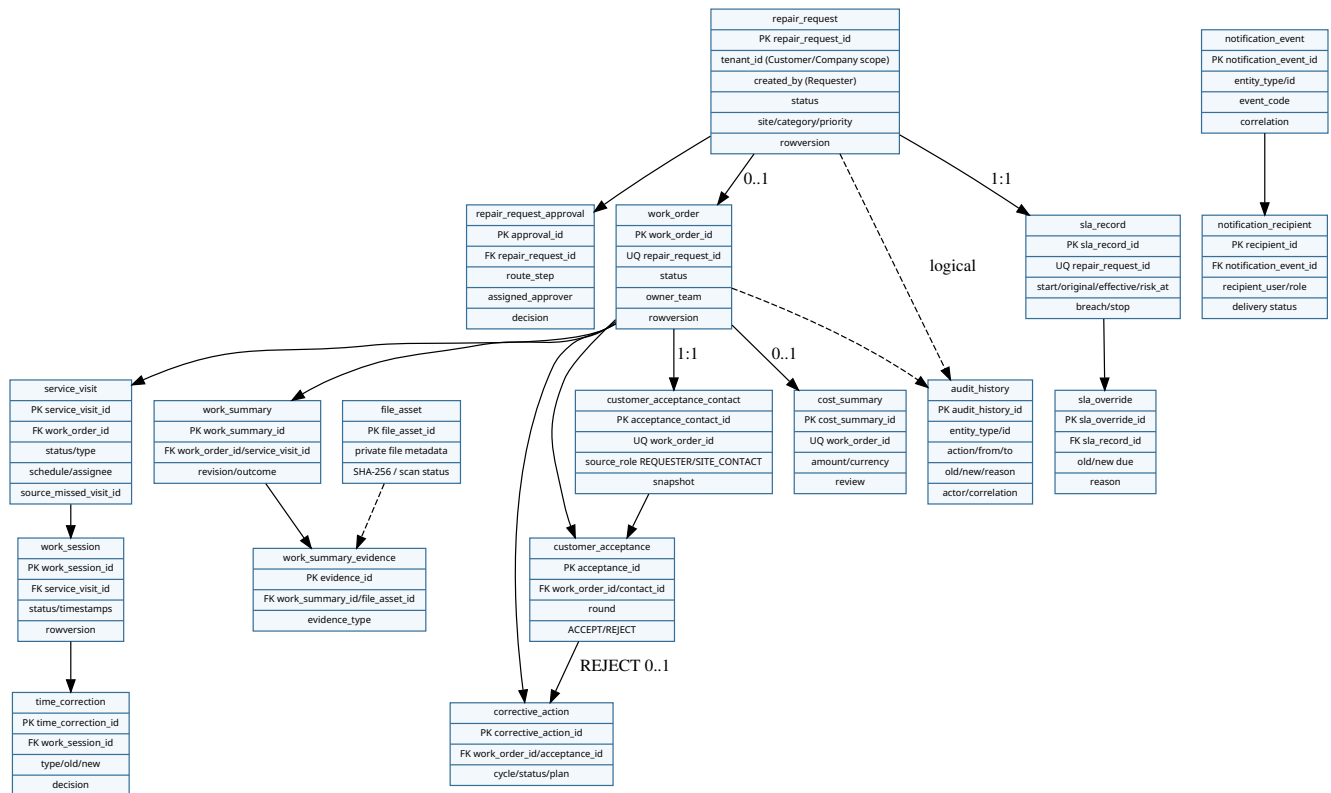
Document Control	Value
Document ID	RR-ERD-001
Version	1.1 - Technical Design Baseline Candidate
Status	Ready for Technical Baseline Approval
Revision Date	12 September 2026
Approval	Business Owner / Project Manager - formal name/signature pending
Sources	Requirement v1.5; BR v1.5; State v1.5; UC v1.6; DD v1.3; Test v1.2
Audience	SA, Database Architect, Backend Developer, QA
Language	ภาษาไทยเป็นหลัก พร้อมชื่อ Entity / Role / State / API เป็นภาษาอังกฤษ
Purpose	แสดง Entity, Cardinality, Ownership and physical mapping boundaries for Repair Request MVP.

Revision rule: เอกสารฉบับนี้ปิด Audit findings ของชุดวันที่ 11 September 2026 โดยแยกสิ่งที่เป็น Business Baseline ออกจาก Technical Design อย่างชัดเจน. การลงนามอนุมัติจริงยังเป็นหน้าที่ของ Business Owner / Project Manager.

1. Design Boundary

ERD maps the approved/candidate business semantics without inventing master-data internal schemas. Tenant/ User/Site/Category/Priority/Team/Outcome/Evidence Type are references only. FileAsset and NotificationRecipient are technical support entities required by file lifecycle and deterministic in-system notification routing.

2. Logical / Physical ER Diagram



Revised ERD v1.1 - corrected acceptance contact, notification recipient and SLA risk fields.

3. Relationship Catalog

Parent	Cardinality	Child	Key / Rule
RepairRequest	1:N	RequestAttachment	request_attachment.repair_request_id
RepairRequest	1:N	ApprovalDecision	approval.repair_request_id
RepairRequest	1:0..1	WorkOrder	UNIQUE work_order.repair_request_id
RepairRequest	1:1	SLARecord	UNIQUE sla_record.repair_request_id
SLARecord	1:N	SLAOverride	sla_override.sla_record_id
WorkOrder	1:N	ServiceVisit	service_visit.work_order_id
ServiceVisit	1:N	WorkSession	work_session.service_visit_id
WorkSession	1:N	TimeCorrection	time_correction.work_session_id
WorkOrder	1:N	WorkSummary	work_summary.work_order_id

Parent	Cardinality	Child	Key / Rule
WorkSummary	1:N	WorkSummaryEvidence	evidence.work_summary_id
FileAsset	1:N	Attachment/Evidence	private shared file metadata
WorkOrder	1:0..1	CostSummary	UNIQUE cost_summary.work_order_id
WorkOrder	1:1	CustomerAcceptanceContact	source role REQUESTER/SITE_CONTACT only
WorkOrder	1:N	CustomerAcceptance	UNIQUE(work_order_id, round_no)
CustomerAcceptance	1:0..1	CorrectiveAction	REJECT creates one corrective cycle
ServiceVisit(MISSED)	1:N	Follow-up ServiceVisit	source_missed_visit_id; original immutable
NotificationEvent	1:N	NotificationRecipient	event recipients / delivery state
Audit / Notification	logical polymorphic	Business entities	entity_type + entity_id; service validates scope

4. Technical Design Decisions

ID	Decision	Reason / Traceability
ERD-DR-01	ApprovalRoute header + ApprovalRouteStep detail	Source route has ordered steps; normalize UNIQUE(route_id, step_no).
ERD-DR-02	Shared FileAsset	Supports attachment, corrective plan and work evidence lifecycle with private metadata/scan status.
ERD-DR-03	Acceptance contact one-way FK from contact to WO	Preserves 1:1 designated contact while avoiding circular FK; role restricted to Requester/Site Contact.
ERD-DR-04	NotificationEvent + NotificationRecipient	D-14 requires deterministic multi-recipient routing/retry without business rollback.
ERD-DR-05	SLA risk_at snapshot	Implements D-13; recomputed on Override; breach remains immutable once set.
ERD-DR-06	Master data references only	Source does not define internal master schemas; do not invent fields.

5. Readiness

ERD is synchronized to Requirement v1.5 / BR v1.5 / State v1.5 / UC v1.6 / DD v1.3 / Test v1.2 and is ready for technical baseline approval.

Performance Audit Revision Appendix

ส่วนนี้เป็น revision content ที่มีผลเหนือ technical note เดิมเฉพาะเมื่อมีข้อขัดแย้งกัน โดยไม่เปลี่ยน Business Baseline.

6. ERD Performance Boundary Review

No business entity, cardinality or relationship is added by this review. Physical access paths remain owned by Database Design.

Relationship / Area	Performance Relevance	Boundary
RepairRequest -> WorkOrder / SLA / Audit	List/detail/report joins and timeline growth	Do not change cardinality; use projected queries and DBD index strategy.
ServiceVisit -> WorkSession / Summary / Evidence	Technician/detail path can grow per WO	Do not eager-load full graph for list screens; EF query shape is Architecture/Implementation concern.
NotificationEvent -> NotificationRecipient	Inbox/retry high-frequency technical path	Event owns created_at; recipient owns delivery status/attempt fields. No recipient.created_at is introduced.
Audit polymorphic reference	Timeline can grow without FK to every business entity	Scope and entity filter must occur server-side before paging; archive/retention remains deployment policy.
FileAsset -> Attachment/ Evidence	Binary growth is external to SQL	DB stores metadata/reference only; private object storage remains authoritative for binary.

7. Physical Design Trace

Index/partition/cache/materialization decisions are not part of the logical ER model. RR-DBD-001 and RR-ARCH-001 may optimize physical access only after representative query-plan/load evidence and must preserve this ERD's ownership/cardinality semantics.